Big-O:

LC) Enc: $O(Th_c(M+T_m)+h_c^2)$

Dec: $O\left(\frac{K}{M}h_c(M+T_m)+h_c^2+h_c 2^m\right)$

BAF) Enc: $O(T_m(L_B^E l h_B^2 + l h_B(T_m+M)))$

Dec: $O(L_B^D l h_B^2 + l h_B(M+T_m) + l h_B^2 + 2^{m+1} l h_B)$

RNN) Enc: $O(Th_r^2+Mh_r)$

Dec: $O\left(\frac{K}{m}(T_m(h_r^2+Mh_r)+2^m h_r)\right)$

"For our setting, FLOPS counts totalled ~600K, ~2.5M, and ~2.6M for TWLC, TWBAF, & TWRNN, resp."

Per single input vector $\vec{x}$. Does ReLU count towards it?

LC) $\vec{x} \in \mathbb{R}^{m+2(T-1)}$. Let $D := d\_model$, $\overline{T} := \lfloor \frac{K}{m} \rfloor$

normalization & pwr reweight

Layer	ENC	DEC
L1	$2D(M+2(T_m-1))\cdot 2$	$2\cdot 2D(M+2T_m)$
L2	$2\cdot 4D^2$	$2\cdot 4D^2$
L3	$2\cdot 4D^2$	$2\cdot 4D^2$
L4	$2\cdot 4D\cdot D$	$2\cdot 4D^2$
MLP 1	$2\cdot D$	$2\cdot D^2$
MLP 2		$2\cdot D\cdot 2^m$
SKIP	$2D$	$2D$
PWR	3	
LN	$2D^2+18D$	$2D^2+18D$
SFTMAX		$3\cdot 2^m-1$ (see RNN DEC)

same calculation as for TWBAF w/ $B=1$

$$\text{ENC: } T_m K/m\left(2D(M+2(T_m-1))\cdot 2+24D^2+4D+2D^2+18D+3\right.$$

$$= T\left(4D(\text{''})+26D^2+22D+3\right)$$

$$\text{DEC} := \frac{K}{m}\left(4D(M+2T_m)+24D^2+2D^2+2D2^m+2D+2D^2+18D+3\cdot 2^m\right.$$

$$\frac{K}{m}\left(4D(M+2T_m)+28D^2+20D+D2^{m+1}+3\cdot 2^m-1\right)$$

give closed-form for this

6BAF) Let B = # blocks (i.e., input matrix rows)
Let D = d_model, D_e = dim_embed, ℓ = # attn layers
Need to calculate for $\hat{i} = 3D$
FE, POSENC, ATTN, MLP

FE:

Layer	ENC	DEC
1	$2 \cdot \ell \cdot (2(T_m - 1) + M) 3D$	$2 \cdot \ell \cdot (M + 2T_M) 3D$
2	$2 \cdot \ell \cdot 9D^2$	$2 \ell \, 9D^2$
3	$2 \cdot \ell \cdot 9D^2$	$2 \ell \, 9D^2$

PE: $B \cdot D$

MLP:

Layer	ENC	DEC
tanh	8ℓ	8ℓ
1	$\cancel{2 \cdot D \cdot \frac{D}{4} \cdot \ell}$	$2 \cdot \frac{D}{4} \cdot 2^m \cdot \ell$
2	$2 \cdot \frac{D}{4} \cdot 1 \cdot \ell$	
PWR	3ℓ	

ATTN: Key (k), Query (q), Val (v): $\ell \cdot D^2, \ell D^2, \ell D^2$
$\vec{k} \cdot \vec{q}$: $\ell \cdot \ell \cdot$ #heads $\cdot$ head_dim
$\vec{attn} \cdot \vec{v}$: $\ell^2 \cdot$ #heads $\cdot$ head_dim

× # ATTN layers

$h1$: $x = x + SA(LN(x))$
$h2$: $x = LN(x + FF(x))$

FF: $L2(L1(x))$
SA: (Prev page)
L1: $2 \cdot \ell \cdot D \cdot 4D$
L2: $2 \cdot \ell \cdot 4D \cdot D$

Set num_heads = 1. Then head_dim = D
SA: $2 \cdot 5 \ell D^2$
LN: $\mathbb{E}[x] \sim 2 \cdot (\ell D)^2$ #add then divide
$\quad Var(x) \sim 2 \cdot 3 \cdot \ell D$
$\quad \sqrt{\cdot + \varepsilon} \sim 2 \cdot 2 \cdot \ell D$
$\quad$ -running mean, ÷denom $\sim 2 \cdot 2 \cdot \ell D$
$+ \quad \times\gamma + \beta \sim 2 \cdot 2 \cdot \ell D$

$2\ell^2 D^2 + 18 \ell D$

- - - - - - - - - - - -

$h1$: $\ell D + 10 \ell D^2 + 18 \ell D$
2 + 8 from SA

$h2$: $10 \ell D^2 + 18 \ell D + \ell D + 8 \ell D^2 + 8 \ell D^2$

ENC: $= \left(\begin{array}{ll} h(36 \ell D^2 + 38 \ell D) & \text{\# ATTN} \\ + \ell D & \text{\# PE} \\ + 6 \ell (2(T_m - 1) + M) D + 36 \ell D^2 & \text{\# FE} \\ + 8\ell \quad + D\ell/2 + 3\ell & \text{\# MLP} \end{array} \right) \frac{T_m}{K}$

DEC: $= L(36 \ell D^2 + 38 \ell D) + \ell D$ #ATTN, PE
$+ 6\ell\left(M + \frac{2T_m}{K}\right)D + 36 \ell D^2$ #FE
$+ 8\ell + 2^{m+1} \ell D / 4$ #MLP
$+ \ell(3 \cdot 2^m - 1)$ #SOFTMAX

RNN) ENC:

GRU 1

update gate: $z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z)$

Let $x \in \mathbb{R}^M$, $W_z \in \mathbb{R}^{N_e \times M}$, $U_z \in \mathbb{R}^{N_e \times N_e}$. Then $z \in \mathbb{R}^{N_e}$ & $h_t \in \mathbb{R}^m$.

FLOPs$(z) \sim$

$$
\begin{aligned}
&Uh \to 2N_e^2 \\
&Wx \to 2(M+1)N_e \\
&Q := Wx + Uh \to N_e \\
&Q + b_z \to N_e \\
+\ &\sigma(\text{''}) \to 4N_e \\
\hline
&N_e(2N_e + 2(M+1) + 6) \\
&= N_e(2N_e + 2M + 8)
\end{aligned}
$$

Similarly, $r_t \sim N_e(2N_e + 2M + 8)$

$h_t \sim 4N_e$

$\tilde{h}_t = \tanh(W_h x_t + U_h(r_t \odot h_{t-1}) + b_h)$

$\Rightarrow$ FLOPs$(\tilde{h}) \sim$

$$
\begin{aligned}
&Q := Wx \to 2(M+1)N_e \\
&R := r \odot h \to N_e \\
&S := UR \to 2N_e^2 \\
&T := Q + S \to N_e \\
+\ &T + b_z \to N_e \\
\hline
&N_e(2N_e + 2(M+1) + 3) \\
&= N_e(2N_e + 2M + 5)
\end{aligned}
$$

$\Rightarrow$

$$= 2N_e(2N_e + 2M + 8) + 4N_e + N_e(2N_e + 2M + 5)$$
$$= 6N_e^2 + 6MN_e + 25N_e$$

GRU2 ~

$$r_t + z_t \sim 2N_e(4N_e + 6)$$
$$h_t \sim 4N_e$$
$$+ \quad \tilde{h}_t \sim N_e(4N_e + 3)$$

$$N_e(8N_e + 12 + 4 + 4N_e + 3)$$
$$= N_e(12N_e + 19)$$

$$= 12N_e^2 + 19N_e$$

ENC Linear ~ $2N_e$

$\tanh(ENC_L) \sim 8$

Normalization ~ 2

$+$ PWR ~ 1

$2N_e + 11$

$$\therefore ENC \sim T(18N_e^2 + 6MN_e + 46N_e + 2N_e + 11)$$
$$= T(18N_e^2 + 6MN_e + 48N_e + 11)$$

GRU1

DEC: Uni-directional:

$$z_t \sim N_d(2N_d + 2(M+2) + 6)$$
$$= N_d(2N_d + 2M + 10)$$
$$r_t \sim \text{''}$$
$$h \sim 4N_d$$
$$+ \quad \tilde{h} \sim N_d(2N_d + 2(M+2) + 3)$$

$$T_m(2N_d(2N_d + 2M + 10) + 4N_d + N_d(2N_d + 2(M+2) + 3))$$
$$= T_m(6N_d^2 + 4MN_d + 20N_d + 4N_d + 2MN_d + 4N_d + 3N_d)$$
$$= T_m(6N_d^2 + 6MN_d + 31N_d)$$

$$\Rightarrow \text{bi-directional} = (12N_d^2 + 12MN_d + 31N_d)T_m$$

$$\text{GRU2} \sim (24N_d^2 + 38N_d)T_m$$
$$\text{ATTN} \sim 2(2T_mN_d)$$
$$= 4T_mN_d$$
$$\text{Output} \sim \overset{\text{linear}}{2(2^m \cdot 2N_d)} + \underbrace{\overset{\text{calc exps}}{2^m} + \overset{\text{add}}{(2^m - 1)} + \overset{\text{divide}}{2^m}}_{\text{softmax}}$$
$$= 2^{m+2}N_d + 3 \cdot 2^m - 1$$

$$\therefore \text{DEC} \sim \frac{K}{m}\big(T_m(36N_d^2 + 12MN_d + 69N_d)$$
$$+ 4T_mN_d + 2^{m+2}N_d + 3 \cdot 2^m - 1\big)$$
$$= \frac{K}{m}\big(T_m(36N_d^2 + 18MN_d + 73N_d)$$
$$+ 2^{m+2}N_d + 3 \cdot 2^m - 1\big)$$

In both ENC & DEC, set

$$N_e = N_d = D$$